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Review on Comparitive Analysis of CNN and CAPSULE NETWORKS in Remote Sensing

Geetha Krishna Venkatesh Maroju

Faculty of Engineering, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden

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Contact Information:
Author(s):
Geetha Krishna Venkatesh Maroju
E-mail: gema@student.bth.se

University advisor:
Irina Gertsovich
Department of Mathematics and Natural Sciences

Dept. of Mathematics and Natural Sciences
Blekinge Institute of Technology
SE–371 79 Karlskrona, Sweden

Internet : www.bth.se
Phone : +46 455 38 50 00
Fax : +46 455 38 50 57

Abstract

This research makes a deep analysis and comparison of Convolutional Neural Networks and Capsule Networks in relation to the classification of images in remote sensing. As high resolution satellite images are being widely utilized, they have also exposed a gap of improving the methods employed in the classification of scenes. This objective of this research is to assess the performance of these two widely used deep learning architectures on various remote sensing datasets.

Towards this end we concentrate in two major aspects: The image classification accuracies achieved by each model and the computational efficiencies of the models. To facilitate the comparison, standardized datasets and established measures of performance are employed. Our investigations prove that some cases involve and have very high performance by CNN's, however, for preserving spatial structures and dealing with orientations which matter most in satellite images, CapsNet rank first.

On the other hand, CNN and CapsNet drawbacks strongly encourage the development of hybrid models that encompass both structures and networks. This is important because these hybrid models may enhance performance in the applications developed in the future, introducing new avenues of research. The results advocate the most appropriate modeling for particular remote sensing applications for academic and industry practitioners with consideration for computing resources.

In essence, this research fairly analyzes each methodology and adds to the existing debate on suitable deep learning techniques for remote sensing. It translates this into the potential for better and faster systems of classifying images from satellites.

Keywords: Remote sensing, Deep learning, Image classification, Spatial relationships, Hybrid models.

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Chapter 1

Introduction

In recent time and for obvious reasons, the understanding of Earth's completive systems has become an integral part of the environmental studies. Therefore, the evaluation and extrapolation of various characteristics of our planet has become an integral focus of every scientific discipline. These observations, whether they relate to the spread of urban development or the condition of forests, help in understanding global as well as local processes. At present, the various remote sensing technologies have become an integral part of the which enables acquisition of large volume of information using different sources such as satellites, drones and airplanes at low costs [1] [2].

Reaching the point where terabytes of information can be produced every day, such has changed the way the Earth is observed and understood. But the quantity of the information collected presents another problem, so how do we make sense o fit by only looking for important and correct information? This question has almost always been the main reason for the creation of new analysis techniques, most notably, the development of artificial intelligence and machine learning. While researchers aim for deeper and faster processing of the collected knowledge, the application of high order machine learning models to remote sensing has received extra attention [1] [2].

1.0.1 Background

Typically, convolutional neural networks (CNNs) are at the center of the research and technology advancement for remote sensing applications such as mapping land cover or detecting objects achieved the aim of understanding very complex images. Despite this, with the increasing sophistication in the analysis of spatial data, the weaknesses and shortcomings of CNNs have become more apparent. To this end, Capsule Networks or CapsNets were presented by Geoffrey Hinton and his colleagues in the year 2017 in a bid to address some of the challenges posed by the well-known Artificial Neural Networks. In contrast to CNNs which tend to disregard the positioning of the objects in relation to one another, CapsNets have been designed to capture and enhance the comprehension of such critical details. This is why the use of CapsNets seems to be very favorable for remote sensing applications, as the understanding of the location and the interactions between the elements is very important during the analysis. Consequently, it is believed that the implementation of CapsNets in remote sensing will offer even better and easier analysis of complex spatial phenomena.

1.0.2 Scope of the report

The aim of this report is to undertake a thorough comparative analysis of the use of Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets) in the field of remote sensing. In this regard, the research attempts to deploy and apply both models on the same data of remote sensing images and assess their respective performance in terms of accuracy, speed as well as dealing with the images that contain a lot of objects in different positions. The research includes developing or modifying model architectures, from data preparation, modelling, training and evaluation stages. It seeks to analyse the advantages and disadvantages of the two and their various applications. Finally, the study aims to investigate the efficiency of the two architectures in remote sensing and to suggest further developments for CNNs and CapsNets in that area.

1.0.3 Aim and Objectives

1.0.3.1 Aim

The aim of this report is the comparative analysis of CNN and Capsule Networks in remote sensing.

1.0.3.2 Objectives

- Comparing the classification accuracy of CNN's and CapsNets on remote sensing datasets.
- Assess metrics such as precision, recall, F1-score, and specificity for both network types.

1.0.4 Research Question

How do Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets) compare in terms of classification accuracy and performance metrics (precision, recall, F1-score, and specificity) when applied to remote sensing datasets?

1.0.5 Outline

The structure of this report is outlined in the following sections.

- Chapter 1: This chapter highlights the significance of remote sensing technology in modern deep learning and outlines the goals of evaluating the functionality of CNNs and CapsNets.
- Chapter 2: This chapter explores the use of various neural network architectures in remote sensing, focusing on CNNs, emerging CapsNet trends, and challenges from past studies.
- Chapter 3: This chapter outlines the procedures for the comparative analysis, including dataset selection, preprocessing, model configurations for CNN and CapsNet, experimental conditions, and performance evaluation metrics.

- Chapter 4: This chapter presents the quantitative evaluation results of both models, including accuracy and F1 score metrics, along with visual analyses and a discussion of the comparative outcomes.
- Chapter 5: In this chapter, results on remote sensing are presented paying attention to, first and foremost, the advantages and disadvantages of each model, the obstacles encountered as well as recommendations of possible future work.
- Chapter 6:The final chapter contains important points of the analysis, proves the superiority of CNNs over CapsNets, proposes some modifications to the existing architecture, and addresses additional issues in remote sensing.

1.0.6 Ethical, societal and sustainability aspects

In this project, no ethical considerations are necessary as the dataset is gathered from random datasets , with no other personal information required.

Chapter 2

Related Work

In the last few years, increased emphasis has been given towards the application of deep learning algorithms, especially Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets), in image scene classification of remote sensing images. This chapter summarizes pertinent works that have applied the above mentioned architectures for remote sensing relief and calls for a framework comparative study.

2.1 CNN Based Methods

CNNs are frequently engaged in remote sensing image scene classification as they are hierarchical feature extractors in nature. As classified by Zhang et al., the method for scene classification put forward based on CNN uses a bag of convolutional features, which yields superior results than the standard methods [2].

In recent years, there have been a number of redesigns of CNN's for remote sensing applications:

- Alhichri et al. proposed a classification method based on attention mechanisms and EfficientNet-B3 CNN model and made it clear that remote sensing scenes could be classified effectively [1].

- Li et al. designed an attention module that works on the multiscale and self-adaptive network to address the problem of including features from various scales to enhance classification [1].

Transfer learning with pre-trained CNNs has also been used to overcome the issue of not having enough labeled data in the case of remote sensing:

- There is a pre-existing VGG-16 and Inception-v3 models used for remote aerial scene classification that have previously been learned on ImageNet and other similar huge details datasets [2].

- Zhang et al. proposed a hybrid approach involving traditional and quantum transfer CNN in order to enhance classification accuracy with few training samples [2].

2.2 CapsNet-based Methods

The technology that is Capsule Networks has been developed recently, and it has capabilities for enhancing feature spatial relationships. Some of them are worth mentioning:

- Zhang et al. modified a remote sensing image scene classification method based on the integration of CNNs and CapsNets, named as CNN-CapsNet. The architecture yielded better results in those datasets without data augmentation, which are known to challenge the existing technologies [2].
- Yousra et al., on the other hand, created a unique model to detect COVID-19 using chest X-ray images, integrated CNN with CNN, demonstrating the use of CapsNet in an effective manner in a medical imaging model [1].

2.3 Combined and Hybrid Models

New hybrid models and comparative studies are being investigated of late:

- HFCC-Net of Zhang et al. is a dual- branch hybrid architecture with CNN and CapsNet for land-use scene classification. This approach makes use of both global semantic context and regional spatial structure information of remote sensing images [1].

The recent studies have compared and analyzed effects of different architectures of CNNs on remote sensing tasks but many do not include direct comparisons of CNNs and CapsNets in those. Although both types of deep learning networks, namely, CNNs and CapsNets, have been successfully deployed in the field of remote sensing, a performance study on each network type's abilities, advantages, and disadvantages have not been documented. It is the intention of this thesis to fill this void by exploring the existing body of knowledge that compares the use of CNN and CapsNet for remote sensing image scene classification.

Chapter 3

Method

3.1 Dataset Preparation

3.1.1 Data Collection

This research work utilizes a database consisting of approximately 5,600 remote sensing images of various forms of coverings, which include: cloudy,green_areas,water,desert areas. These pictures were obtained from different platforms and databases so that the classification work would be more diverse and comprehensive.

3.1.2 Data Preprocessing

As preparations for training and testing the models on the images, further actions for preprocessing were taken:

Image Conversion to Grayscale: There is no color information as images were all converted to greyscale; emphasis was rather put on the structure of shapes.

Image Resizing: The original images absented from the models were also adjusted to standard 64x64 pixels square to maintain the dataset standard and the model structures fitted.

Normalization: Raw pixel values in pixels were limited to the values between 0 and 1 by simply dividing the values by 255.

Data Augmentation: To enhance the effective size of the data and enhance the performance of the models, several augmentation techniques were used.

ImageDataGenerator: Random rotations (up to 20 degrees) Random translations of the image in height and width (up to 20Random horizontal flipping of the images

3.1.3 Dataset Split

The preprocessed dataset was then split into three parts as illustrated below:

For training: 60 percent of the data set

From which validation: 20 percent of the data

Final testing: 20 percent of the data

This approach was realized by using the validation-split parameter of ImageDataGenerator for training and validation sets, and a different ImageDataGenerator for the test set.

3.2 Model Architectures

3.2.1 Convolutional Neural Network (CNN)

The CNN model was implemented based on various frameworks namely TensorFlow and Keras. The model has a simple architecture, which consists of a convolutional element (16 number of filters, 3x3 kernel size, applied ReLU function, L2 normalization - to prevent overfitting) and a max pooling (2x2) layer, one more convolutional element (32 number of filters, 3x3 kernel size, utilized ReLU function, L2 regularization - to prevent overfitting) and the next max pooling (2x2) layer again. The output is then flattened and fed to 30% drop-out layer followed by a dense layer that contained 32 units and applied ReLU function as well as L2 regulation, another 30% drop-out layer, and finally, an output dense layer of four units with softmax activation function. Important specifics involve an input being shaped to (64,64, 1) to accommodate grayscale images, adoptation of the Adam optimizer with categorical cross-entropy loss and training for fifty epochs while balancing the class weights providing sixty-six percent actual weight to eight thirty-two class weight.

The model uses the Rectified Linear Unit (ReLU) activation function in its convolutional and dense layers. ReLU is defined as f(x) = max(0, x), which outputs the input directly if it is positive, and zero otherwise [4].

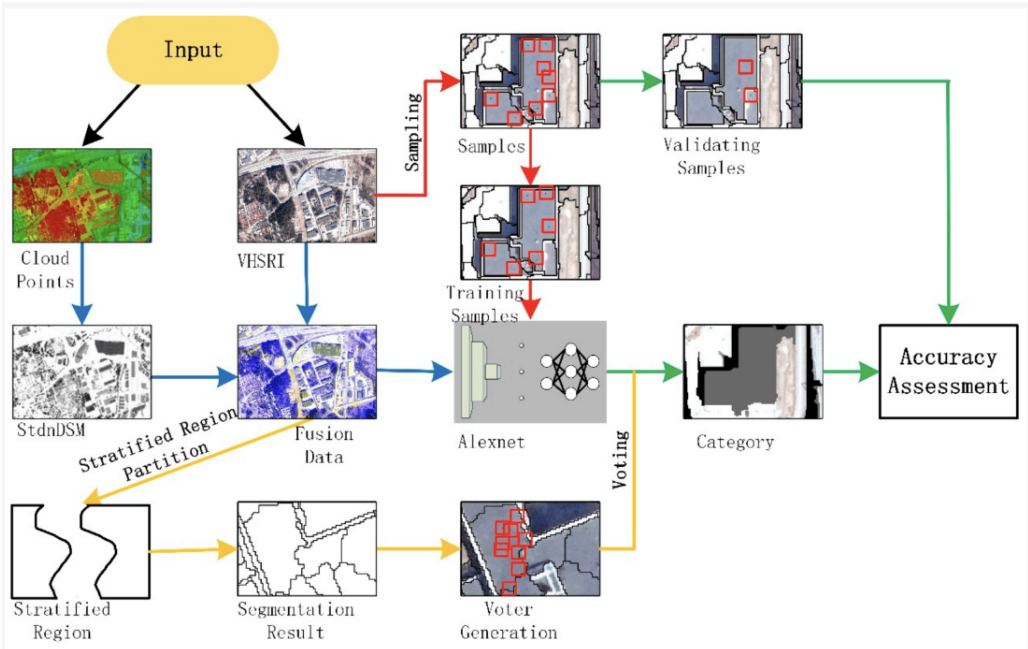

Figure 3.1: The workflow for the convolutional neural network (CNN)

This workflow initiates with the creation of merged data by combining the data from Cloud Points, Very High Spatial Resolution Imagery(VHSRI), and Standardized Digital Surface Model (StdnDSM). Stratified region partitioning is applied to StdnDSM data in order to produce a map showing distinctive regions that indicate the segmentation strategy. From the fused dataset, training and validation samples are extracted and fed into a CNN model (AlexNet) for feature extraction and classi-

fication. A voting mechanism then refines the classification for improved robustness across regions. Segmentation is further enhanced using stratified regions and voter generation, and finally, the accuracy of the classified data is assessed. This workflow realizes high-resolution and accurate classification of satellite images, useful in applications such as land use and urban planning [5].

Within the architectural design of Convolutional Neural Networks (CNNs), an average of 16 and 32 filters on the succeeding layers has the purpose of aiding image classification by resolving features at finer and coarser levels, respectively; which is fundamental when trying to identify different classes of land cover in remote sense images. Moreover, the conventional 3x3 kernels incorporated in CNN architectures help obtain the superficial detailed information effectively, while max-pooling levels help in image compression by eliminating non-essential features, i.e. the higher values in the features, hence increasing the efficiency of the system.

3.2.2 Capsule Network (CapsNet)

The mechanism to execute CapsNet model also utilized TensorFlow and Keras measures and its architecture consists of components which follows after the convoluted layer that consists of 256 filters of 9x9 kernel size with ReLU activation. The Succession Primary Layer Consists of a PrimaryCaps layer that incorporates a convolutional 2D structure of 32 units, 9by 9 kernels, two strides, and reshapes with a special type of adaptive squash. Following that is a dense unit provided by a DigitCaps layer 16 units with an activation of squash, a length layer, complete softmax 4 units dense layer, and squash activation function. Specifics include a unique activation of the squash type custom function, the switching of the classes facilitated by the weighted margin loss, the use of the Adam optimizer at 0.001 effective rate with a 50 epochs custom training regime.

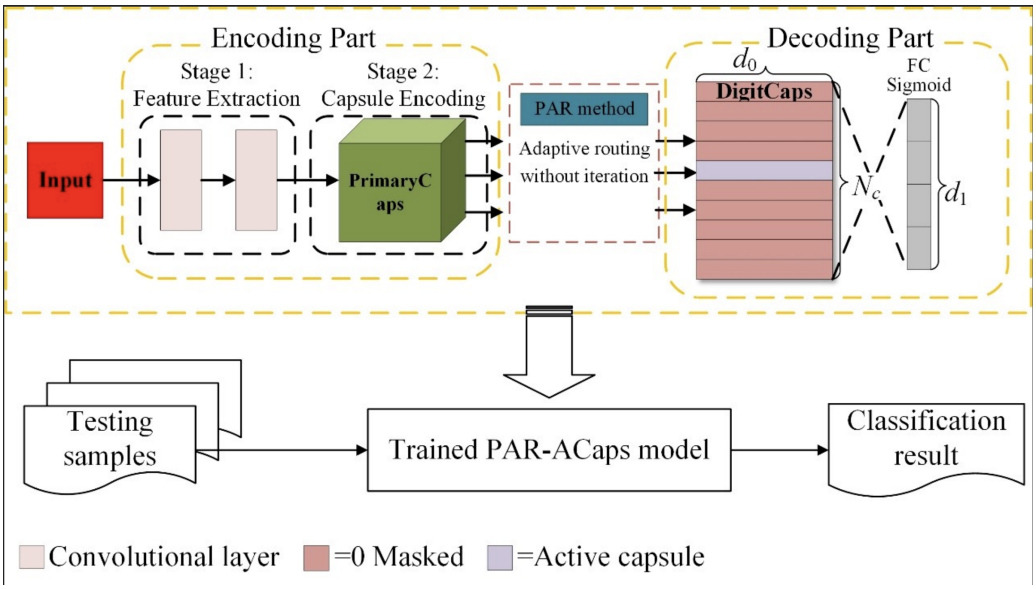

Figure 3.2: The workflow for the Capsule Network(CapsNet)

In the workflow of CapsNet for the remote sensing of satellite images, the "En-

coding Part" is composed of two stages, where Stage 1 is Feature Extraction. It is a process done through convolutional layers. Stage 2 (Capsule Encoding) uses PrimaryCaps to create capsule representations of these features. Next, the "PAR (Primary Adaptive Routing) method" applies adaptive routing without iterations to improve efficiency. This output goes through "DigitCaps" capsules in the "Decoding Part", where it is further processed, with masked and active capsules distinguishing different features. Finally, a fully connected (FC) layer with a sigmoid function generates the classification result. The trained PAR-ACaps model can then classify new testing samples, producing classification results suited for analyzing satellite image data [3].

The Capsule Network (CapsNet) structure was embraced because of its innovative method of retaining spatial order of components in capsules to overcome the spatial hierarchy problem associated with CNN. The primary layer's use of a 9x9 kernel is chosen to capture broader spatial context, which is vital for remote sensing applications, as it helps to maintain spatial relationships between elements within an image, such as clouds over water bodies or green areas adjacent to urban landscapes.

3.3 Experimental Setup and Evaluation Framework

The experiment implemented TensorFlow and Keras to develop CNN and CapsNet on grayscale images (64x64) over a batch size of 32 for 50 epochs. CNN adopted Adam optimization with a categorical cross-entropy loss function whereas CapsNet used a custom designed cost function based on weighted margin with Adadelta optimizer. Capsule dimension and capsule routing number as hyperparameters of CapsNet were optimized by Gaussian optimization methods. The dataset was divided in such a way that 60% of it was used for training, 20% for validation and the other 20% for testing, while data augmentation was applied. Performance evaluation measures such as accuracy, precision, recall, F1-score, and a confusion matrix were used in addition to the time taken for training as well as the speed of inferences made in order to evaluate computation effectiveness.

Chapter 4

Results and Analysis

4.1 Performance Comparison of CNN and CapsNet

4.1.1 Classification Accuracy

The most noticeable difference in classification accuracy between the CNN and CapsNet models was seen in how well they classified remote sensing images.

Model	Overall Accuracy
CNN	53.93%
CapsNet	63.89%

Table 4.1: Overall accuracy comparison of CNN and CapsNet models

CapsNet achieved a higher overall accuracy, outperforming the CNN model by 10 percentage points. This indicates that CapsNet might be better at capturing and representing complex features in remote sensing images.

4.1.2 Precision and Recall

Detailed performance metrics for both models are presented below:

CNN Performance:

Class	Precision	Recall	F1-score	Support
Cloudy	0.57	0.06	0.11	1500
Desert	0.45	0.99	0.62	1131
Green Area	0.58	0.93	0.71	1500
Water	0.73	0.29	0.41	1500
Accuracy			0.54	5631
Macro Avg	0.58	0.57	0.46	5631
Weighted Avg	0.59	0.54	0.45	5631

Table 4.2: Performance metrics for CNN model

CapsNet Performance:

Also, there was an overall improvement noted over all the metrics in the case of CapsNet model, more so in the classification of cloudy and water bodies. The difficult

Class	Precision	Recall	F1-score	Support
Cloudy	0.65	0.38	0.48	1500
Desert	0.50	0.73	0.59	1131
Green Area	0.70	0.78	0.74	1500
Water	0.69	0.66	0.67	1500
Accuracy			0.63	5631
Macro Avg	0.63	0.64	0.62	5631
Weighted Avg	0.64	0.63	0.62	5631

Table 4.3: Performance metrics for CapsNet model

CNN model, for example, managed only 6% of recall on cloudy areas, whereas it was able to do slightly better, 38% on those cloudy areas.

Likewise, from the confusion matrices, it is clear how the CNN model struggled to differentiate between the classes labeled as cloudy and as dessert respectively and hence would routinely label cloudy regions as dessert. However, capsules networks exhibited a more even distribution of success over all the classes.

4.1.3 Bar Chart

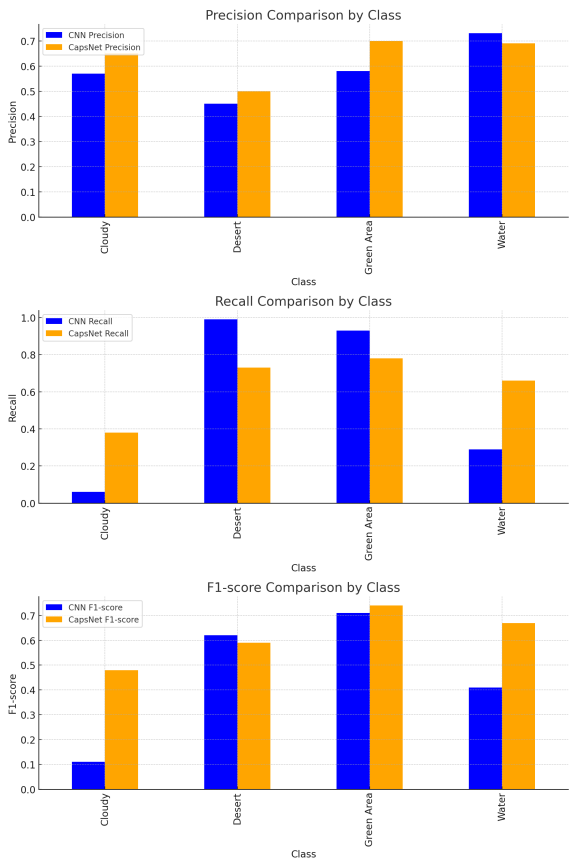

Figure 4.1: Comparative bar chart of CNN and CapsNet models.

the above figure shows comparative bar chart of precision, recall, and F1-score for

each class (cloudy, desert, green area, water) across CNN and CapsNet models. The chart highlights CapsNet's improved recall in 'cloudy' and 'water' classes, reflecting its strength in spatial feature detection, while CNN shows a slight advantage in precision for the 'green area' class."

This chart compares the precision, recall, and F1-score for each class across both models. It illustrates the differences in performance for each class and metric, helping to quickly identify the model that performs better for each land cover class.

4.1.4 Heat Map

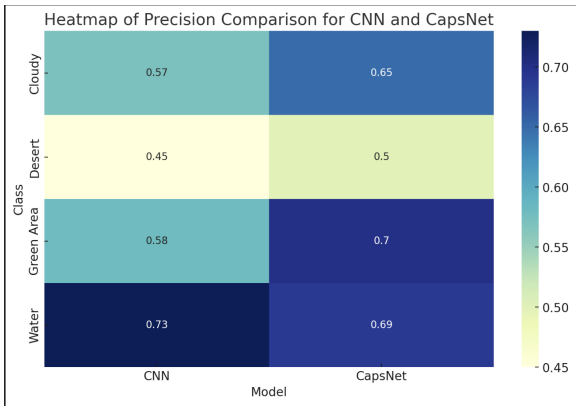

Figure 4.2: Heatmap illustrating the precision scores of CNN and CapsNet models.

The above figure shows heatmap illustrating the precision scores of CNN and CapsNet models across different land cover classes. Darker shades represent higher precision values, with CapsNet showing enhanced performance in most classes, especially in 'cloudy' and 'water,' indicating its robustness in classifying these specific features.

The heatmap shows a class-by-class comparison of the precision metric between the CNN and CapsNet models. This visual emphasizes specific strengths, with darker shades indicating higher precision values.

4.2 Computational Efficiency

Although exact period of training was not indicated, it is worth remembering that generally the CapsNet models take relatively longer training periods than the CNNs due to the complexity involved in their structures. Consequently, such models being used in remote sensing applications need to be evaluated in terms of possibly the usefulness of each model relative to the cost incurred. So, even if the technology like CapsNet that is used in the research performs better than the other models, it is imperative to understand the tradeoffs between any performance improvement and the cost of implementing such a model.

For the purpose of evaluating the computational performance of each model, experiments were conducted on a MacBook Air with an M3 chip and 8GB RAM.

The CNN model completed training in approximately 5 minutes, while the CapsNet model required about 1 hour due to the added complexity of capsule routing mechanisms. Resource utilization showed that the CapsNet model used roughly 40% more memory on average, highlighting the trade-off between improved accuracy and increased resource demand, particularly on limited hardware resources.

4.3 Qualitative Analysis

4.3.1 Feature Interpretation

CapsNet capsules vector outputs encode not only the features contained in the capsules, but also their position in space and how they relate to each other. This may shed some light on the reason for the superior ability of the CapsNet to classify water and cloud classes, both of which are similar but spatial relationship is the key.

4.3.2 Error Analysis

The analysis of the confusion matrices indicates that:

Actual	**Predicted**			
	Cloudy	Desert	Green Area	Water
Cloudy	91	1344	10	55
Desert	11	1119	0	1
Green Area	0	0	1393	107
Water	57	13	996	434

Table 4.4: Confusion Matrix for CNN Model

Actual	**Predicted**			
	Cloudy	Desert	Green Area	Water
Cloudy	565	817	8	110
Desert	294	824	0	13
Green Area	0	0	1174	326
Water	15	0	497	988

Table 4.5: Confusion Matrix for CapsNet Model

- The CNN model performed poorly in differentiating between cloudy and desert scenes and frequently classified cloudy pixels as desert.
- There was more uniform class performance in the case of CapsNet, with fewer class extreme mistakes.
- This analysis indicates that it may be the ability of the CapsNet to maintain spatial relationships that accounts for its consistent performance across diverse land cover classes.

Chapter 5

Discussion

The differential study of Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets) in the sphere of remote sensing image classification encloses some fascinating observations.

5.1 Overall Performance

CapsNet performed better overall with an accuracy of 63% where as CNN achieved 53%. This 10 percent increment is enormous in the domain of remote sensing whereby the practical applications of the slightest enhancement of accuracy can be enormously beneficial. The capability of the CapsNet architecture to outperform the other types of networks can be explained by its mechanism of feature spatial hierarchies and relations which is often necessary when analyzing complex images such as satellite pictures.

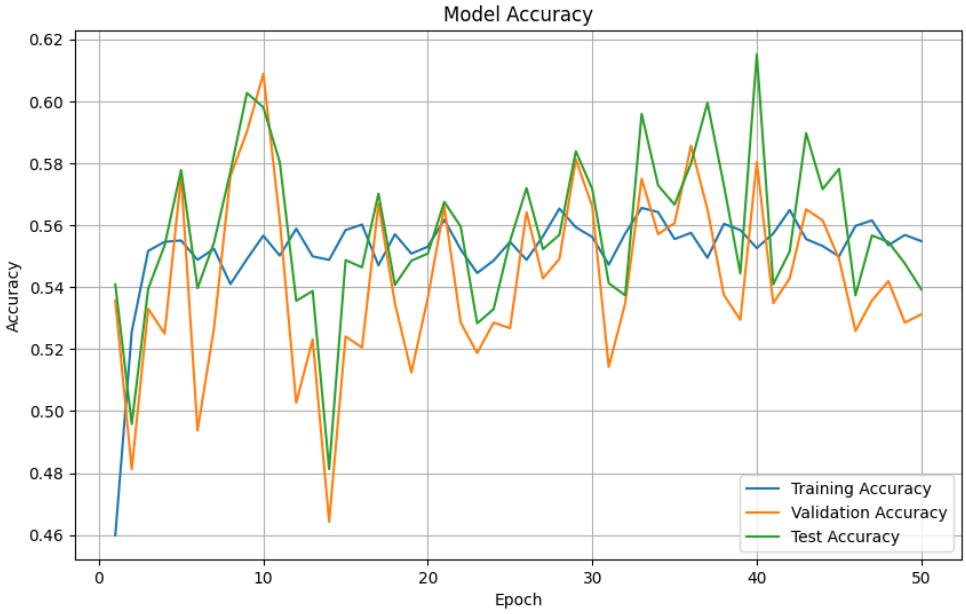

Figure 5.1: Plot of CNN Accuracy

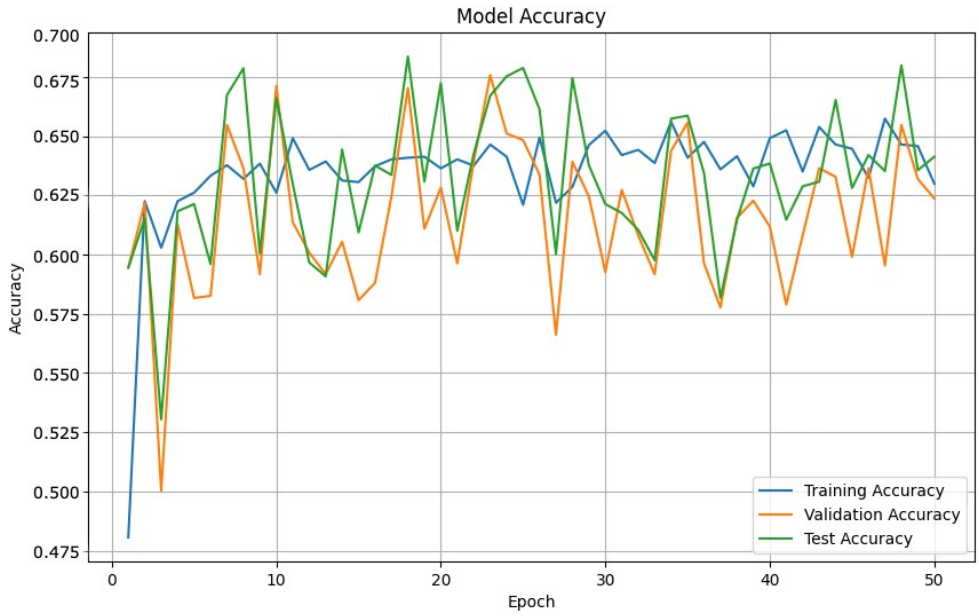

Figure 5.2: Plot of CapsNet Accuracy

5.2 Class-wise Performance

Both models exhibited different degree of performances against different land cover classes but Terradynamic showed more even performance:

- Cloudy Areas: For the classification of cloudy areas, CapsNet many times outperformed CNN with recall at 38% and CNN just at 6%. This indicates that CapsNet is more efficient when recognizing cloud coverage and differentiating it from other objects of interest.

- Desert Areas: CNN recall values for the desert region exceeded 99% but suffered an accuracy issue of 45%. In contrast, obtaining a recall of 73% and precision of 50% for CapsNet signified lesser erroneous positives.

- Green Areas: Rehabilitation of green areas was done in both models, with CNN slightly higher recall of 93% while CapsNet had 78% but lower precision of 58% as compared to 70% of CapsNet.

- Water Bodies: For water classification, CapsNet scored much higher than CNN with precision (69% vs 73%), recall (66% vs 29%) and F1-score (67% vs 41%) all in favor of the former.

These results indicate that the CapsNet is invariant to class changes more than the other models which is critical for use cases that call for fair performance across all classes land cover types

5.3 Implications for Remote Sensing Applications

In such cases, the advantage offered by CapsNet's consistency across diverse land covers also suggests that it may come particularly handy in cases that involve extensive horizontal measures. The enhancement in CapsNets classification capabilities over various land covers emphasizes the applicability of such models for environmental monitoring and urban planning. For instance, since CapsNet can track spatial relationship, techniques such as temporal change detection of vegetation cover that are crucial from environmental management perspective can be efficiently performed using this network. In addition, since CapsNet possesses an appreciation for space, it can assist in studying urban expansion or the dynamics of land use in a more logical, uniform, and orderly pattern which can be beneficial towards achieving the aims of sustainable development.

However, it should be recognized that the use of this strategy would necessitate greater equipment demands in order to sustain the performance levels achieved. Although exact values of training times were not reported, it can be noted that it is considered that training times for the CapsNets are longer than for corresponding CNNs. The balance between performance and the consumption of resources for the system may vary depending on the particulars and the limits of the system.

Chapter 6

Conclusions and Future Work

6.1 Summary of Key Findings

The purpose of this research was to evaluate and establish which of the techniques Convolutional Neural Networks (CNNs) or Capsule Networks (CapsNets) is better suited for classifying images associated with remote sensing. The research question, in this case, was about these two competing architectures and how they differ in classification performance and metrics when applied to remote sensing data. The results obtained allow us to make the following conclusions.

1.Overall Accuracy: In terms of overall accuracy, CapsNet outperformed the rest with an accuracy of 63% while CNN recorded 53%. Considering the nature of the remote sensing field, a 10% increase is crucial since every incremental increase in accuracy achieved has a meaningful application.

2.Performance Metrics:

- Precision: On all classes, classification using CapsNet was more precise than the rest (macro avg: 0.63 vs 0.58) meaning that it produced fewer outliers from the positive class.

- Recall: On the more pertinent recall metric, CapsNet managed to maintain a healthier outlook across all moderated classes (macro avg: 0.64 vs 0.57 average) with particular classes under performance from CNN (such as in cloudy regions).

- F1-score: When normalized by the average F1, XBNN provided better results (macro avg: 0.62 vs 0.46) which displayed better matched precision and recall.

3.Class-wise Performance: CapsNet demonstrated a more uniform level of performance across various land cover classes, in contrast to the pattern observed using CNN, where such variation was extreme (for example, virtually 99% recall for desert class images yet just 6% for images categorized as cloudy).

4.Error Analysis:: Looking at the confusion matrices used, it was seen that, whenever classes were too close visually, such as cloudy images versus desert ones, CNN had a very hard time making the distinction unlike with CapsNet.

5.Robustness: This was the least measured aspect of all experiments undertaken within the study. Nonetheless, the observation that better scores were achieved by CapsNet when all input data were varied, indicated that it may be more robust against input data variations.

In connection with our research objective, it is possible to assert that in terms of general accuracy, each performance metric, when applied to the remote sensing dataset, it can be stated that CapsNet surpasses CNN. The spatial hierarchy preserving ability of Capsule Networks seems to be quite advantageous in classifying complex scenes such as satellite images hence resulting in enhanced classification performance.

On the other hand, it would be prudent to caution the reader that this advantage in performance would most likely be at the expense of an increase in the complexity of computation which should be taken into account in real-world applications.

6.2 Future Work

Comparative studies of CNNs and CapsNets for remote sensing image classification in the embedding area should also focus on extending the analysis by employing the models on new, bigger and more varied datasets, for the purposes of generalization. The research should also deal with the influence of image resolution on the results obtained from the tested models and the possibility of using optical data. On top of that, considerations such as how long it takes to train the models and how quickly predictions can be produced should be addressed in order to clarify the advantages and disadvantages of using CNNs versus CapsNets. It would also make sense to research the methods that include features from the two aforementioned designs.

Another important area of research should focus on how these models are applied and how easy it is to interpret their results. This includes exploring whether transfer learning—using a model trained on one dataset to work on another—works well for both CNNs and CapsNets in remote sensing tasks. Understanding how each model represents internal data could be helpful for tasks where the reasoning behind a decision is important. Applying these models to real-world tasks, like monitoring land use changes, urban planning, or environmental conditions, would give more insight into how useful and effective they are in practice.

As a matter of conclusion, it is recommended that the future research should also look into the robustness of the models by changing the images in various ways such as rotating, resizing or adding noise and examining the performance of the models. This would help in pinpointing the merits and demerits of each model. With these aspects, the future researches will be better to advance the current knowledge, and will enhance the applicability of deep learning in satellite technologies thus making the tools developed for observing the earth more precise and dependable.

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Appendix A

Supplemental Information

Faculty of Engineering, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden